

Project 08: Visual QA & Image Retrieval Studio

Multimodal Vision-Language Inspection Engine

1. Executive Overview

The Visual QA Studio combines image metadata retrieval with multimodal question answering. It indexes visual assets by tag, description, and OCR text, retrieves matching catalog images, and synthesizes answers based on visual content.

2. System Architecture & Workflow Pipeline

- Catalog Image Keyword Indexer (Tags, Descriptions, OCR Text)
- Top-K Asset Retrieval & Metadata Extraction
- Google Gemini 2.5 Flash Multimodal VLM Grounded Synthesis
- Interactive Dashboard displaying image previews, tags, and OCR overlays

3. API Endpoints Specification

Method	Endpoint	Description
GET	/	Serves Visual QA Studio web UI
POST	/api/visual-qa	Executes visual retrieval and returns multimodal answer

4. Deployment & Automated Verification

Standalone Uvicorn Launch:

```
cd 08_visual_qa/src && python -m uvicorn server:app --host 127.0.0.1 --port 8008
```

Automated Test Suite Execution:

```
python -m pytest 08_visual_qa/tests/
```